VATAVARAN

Weather, Your Way

A web-based weather dashboard

Internship Project Presentation



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Introduction

Vatavaran is a web-based weather dashboard that brings location search, current weather information and search history together in a single application.

It combines a React.js frontend, a Node.js and Express.js backend, the Open-Meteo APIs and a MongoDB database to present weather details in a clear, easy-to-understand format.

3 Team members

2 Open-Meteo APIs

1 MongoDB database

Problem Statement



The Need

Users want quick weather details for any city or for their own location.



The Challenge

Weather data must be fetched from external services and shown in a format that is simple to read.



Our Solution

Vatavaran combines search, geolocation, clear weather details and saved search history in one dashboard.

Project Objectives

- Provide real-time current weather information.
- Enable city-based weather searches.
- Support location-based weather detection.
- Display weather details in an easy-to-understand format.
- Provide a responsive and user-friendly interface.
- Store previously searched locations for convenient access.

Features of Vatavaran

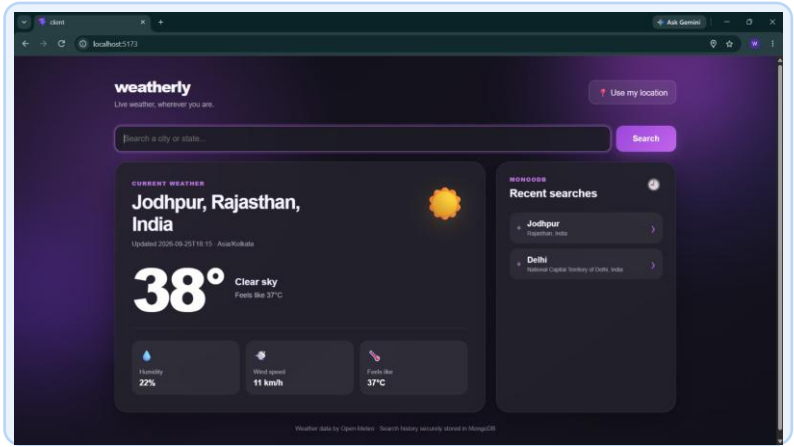
Current Weather
Temperature, humidity, apparent temperature, wind speed and conditions.

Location Search
Search cities and locations to retrieve their weather.

Geolocation
Get weather using geographical coordinates.

Search History
Keep a record of searched locations for future reference.

Responsive Interface
User-friendly layout across different screen sizes.

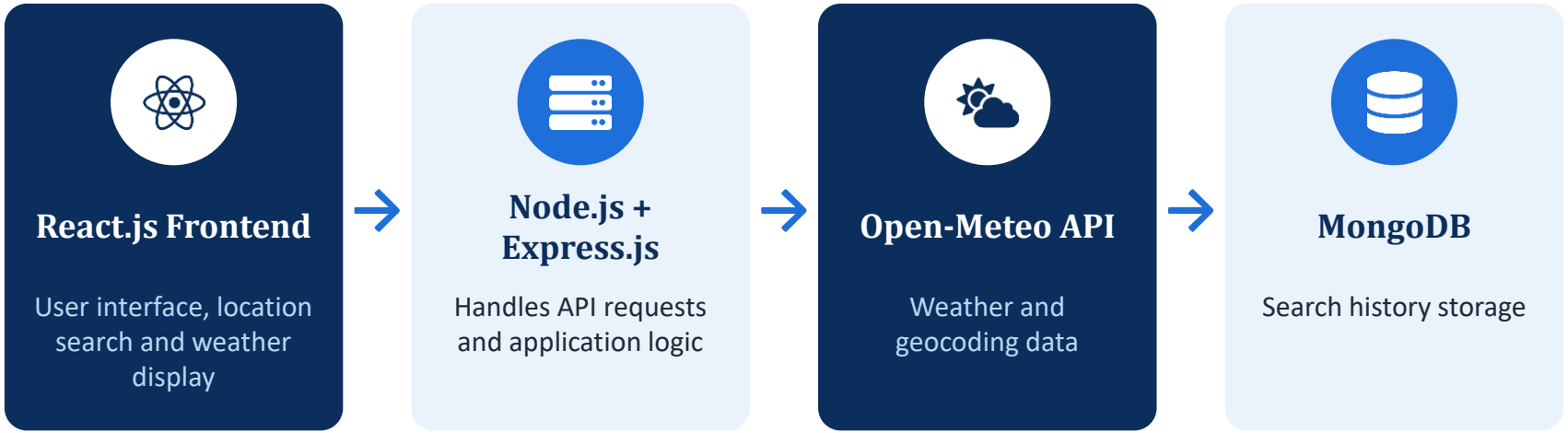


Application screenshot

Technologies Used

Technology	Purpose
React.js	Frontend development
Vite	Development and build tool
Node.js	Backend runtime
Express.js	Backend API and routing
MongoDB	Search history storage
Open-Meteo API	Weather information
Open-Meteo Geocoding API	Location search
HTML, CSS, JavaScript	Web interface development

How the System Works



The backend processes the data and returns results to the frontend.

API Integration and Database

API Integration

- Open-Meteo Weather API: retrieves current weather information.
- Open-Meteo Geocoding API: finds coordinates and location details from search queries.
- Location API: retrieves weather using latitude and longitude.

Database — MongoDB

- City name
- State
- Country
- Latitude and longitude
- Search date and time

Stored search details give the app a history of searched locations.

User Interface and Future Scope

User Interface

- Clean and modern weather dashboard.
- Easy-to-use location search.
- Clear presentation of weather details.
- Responsive layout for different devices.

Future Scope

- Multi-day weather forecasts.
- Weather alerts and notifications.
- Weather charts and visual analytics.
- Improved location detection.
- Personalized user accounts and settings.



UI design concept: light and dark themes, desktop and mobile

Future scope features are planned enhancements, not necessarily part of the current implementation.

Our Team and Conclusion

Lotwan Amarjeet Kaur
UI/UX Design

Dipak Yadav
Frontend Development

Vishal Sukhnani
Backend Development

Conclusion

Vatavaran is a web-based weather dashboard that brings together location search, current weather information and search history in a single application.

The project demonstrates the integration of frontend and backend technologies with external APIs and a database to provide an accessible weather information platform.

Thank You! Vatavaran — Weather, Your Way